

WASP-2b

R-2385

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_120611 · created 2026-08-20T23:07:16+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456089.93027 +/- 0.00078 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1604 +/- 0.0022
Transit depth (Rp/Rs)^2	2.57 +/- 0.069 [%]
Orbital Inclination (inc)	87.26 +/- 0.39
Airmass coefficient 1 (a1)	0.834 +/- 0.015
Airmass coefficient 2 (a2)	0.164 +/- 0.051
Scatter in the residuals of the lightcurve fit is	1.77 %
Best Comparison Star	#2 - [101, 60]
Optimal Aperture	3.65
Optimal Annulus	12.73
Transit Duration (day)	0.0853 +/- 0.0011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1604 ± 0.0022 vs 0.1326 (deviation 3.03σ)
Duration fit vs published	0.0853 d vs 0.0737 d (deviation 2.52σ)
Mid-time O–C	-2.2 min
Depth SNR	17.5
Residual scatter	1.77 %

Light curve

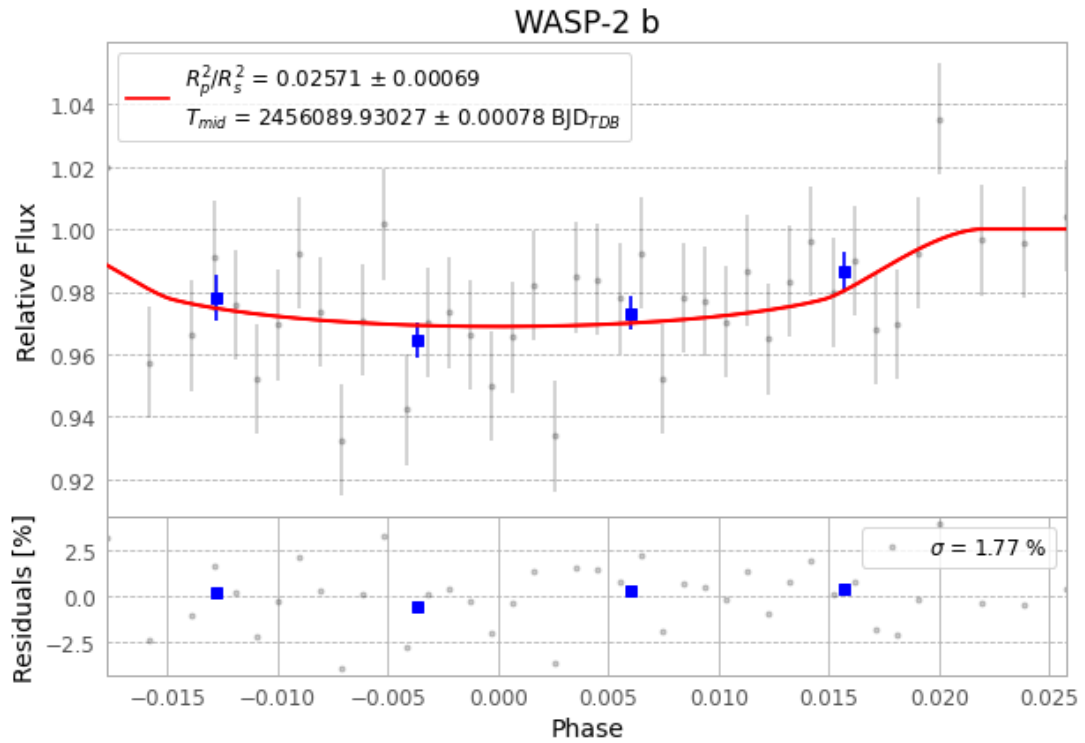


Figure 1 — FinalLightCurve_WASP-2 b_2012-06-11.png (SHA-256 878fcb02a612fa5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [355, 265], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9384ce098f7f235f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

45 FITS frames (30 MB), WASP-2120611091813.FITS → WASP-2120611114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-120611.log (53badb3ac22c...), FinalLightCurve_WASP-2 b_2012-06-11.png (878fcb02a61...), FinalLightCurve_WASP-2 b_2012-06-11.csv (e10739aae7c7...)

Compute resources

Wall time 122.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-20 23:07Z.